An EEG-Based Seizure Prediction Model Encoding Brain Network Temporal Dynamics

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***Abstract*—EEG-based seizure prediction enables timely treatment for patients, but its performance is limited by the difficulty in effectively characterizing the temporal dynamics of epileptic brain networks. Metastability, which describes recurring topographical patterns of spontaneous neural activity over time, provides a unique perspective for capturing the dynamic evolution before seizure onset. In this study, we propose a seizure prediction model that fuses consistent epileptic network processes across subjects into a higher-order latent space. Specifically, we first construct metastable transition patterns to identify the recurrent network states over time. Through adversarial feature learning, we then impose the metastability prior on the latent embedding space encoded via a variational autoencoder (VAE), while leveraging the maximum mean discrepancy measure (MMD) to further mitigate the patient gap. The latent representation, endowed with physiological priors, is ultimately utilized for patient-independent seizure prediction. We evaluate our method on two publicly available and one clinical scalp EEG datasets. Compared to the existing methods, our method has improved AUC, sensitivity, and specificity on CHB-MIT dataset by approximately 9%, 5%, and 5%, respectively. Our method shows that combining brain network-based physiological prior with deep learning for EEG representation learning is a brand-new strategy for associating seizures with complex brain network variations, enabling reliable patient-independent seizure prediction.**

***Index Terms*—Adversarial inference, electroencephalogram (EEG), metastability analysis, seizure prediction.**

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I. Introduction

EPILEPSY is a disorder of brain networks caused by abnormal, hyper-synchronous, and paroxysmal neuronal activity [1], [2]. The scalp electroencephalogram (EEG) is one of the most versatile tools for clinical diagnosis and retrospective analysis of epilepsy [3]. In this case, effectively utilizing EEG data to characterize the evolution of epileptic networks is crucial for accurate seizure prediction.

Considerable research has demonstrated the feasibility of modeling epileptic dynamic transitions from a global network perspective [4], [5], [6]. For example, altered patterns of β- and γ-band brain network connectivity are potential biomarkers for monitoring seizure cycles in West syndrome [7]. The interictal suppression hypothesis in focal epilepsy has gained network-level supporting evidence [8]. However, traditional statistical tools or machine learning used in these approaches are often deficient in modeling high order nonlinear intrinsic, limiting their performance. Graph convolutional network (GCN) models, which capture the structural information within feature interactions in the graph domain, have been widely used for associating seizure disorders with complex brain network variations [9], [10], [11]. Given that seizure-related brain regions are not stably uniformly distributed in the whole-brain network, conventional graph learning approaches, mostly based on fixed graphs, are incapable of fully modeling the dynamics transition during the seizure-related periods.

Originally attributed to constant coupling and decoupling of neural populations [12], metastability serves as a mechanism to describe the dynamic, self-organizing neural process [13], [14], [15], [16]. The metastable substates are transient, quasi-stable

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neural coupling patterns that recur sequentially over time within EEG recordings. It has been utilized to explore the dynamic brain states during anesthesia [17], as well as neurodegenerative diseases modulated by temporal dependencies, such as amyotrophic lateral sclerosis [18] and Alzheimer's disease [19]. Metastability analysis has the potential to provide new insights into the temporal dynamics within functional brain networks in epilepsy.

Integrating established domain knowledge with data-driven deep learning, known as knowledge-augmented [20] and model-based deep learning [21] paradigms, is one of an increasing focus in artificial intelligence, aiming to create more robust and interpretable models. In patient-independent seizure prediction, existing studies have attempted to introduce knowledge priors based on general statistical properties like the Laplace distribution [22], or the geometric structures of EEG signal [23], to overcome the significant variability and noise in EEG signals. Inspired by the principles of incorporating domain knowledge to guide data-driven models, in this study, we propose a patient-independent seizure prediction model, MaDin, which explicitly encodes the metastable properties of epileptic networks, a physiologically-grounded prior, within a deep learning framework. Specifically, this work introduces a novel prior based on metastability analysis. To preserve higher-order nonlinear information and generalize better on the target patient, a variational autoencoder (VAE) is then employed to learn a latent embedding space that approximates a normal distribution. The aggregated posterior of the latent space is matched with the metastability prior through adversarial feature learning, while the maximum mean discrepancy (MMD) measure is used to align the distribution across subjects. This approach ensures that the learned latent representation captures both subject-invariant information and disease-related neural features, enabling reliable patient-independent seizure prediction. We have evaluated our method on CHB-MIT [24], Siena epilepsy datasets [25], and a clinical dataset. Our contributions are summarized below:

1) We propose a patient-independent seizure prediction model, MaDin, that fuses metastability prior to latent representations through adversarial inference. The learnt epileptic representations are encoded with brain network temporal dynamics and sufficient in high-order nonlinear intrinsic attributes.
2) For the first time, we characterize metastable properties to investigate the temporal dynamics of brain networks during seizure evolution. Consistent metastable patterns across patients reflect common epileptic network behaviors, which are key to achieving reliable prediction performance.
3) Our MaDin demonstrates promising prediction performance on two publicly available and one clinical scalp EEG datasets, outperforming existing patient-independent models on the CHB-MIT dataset.

II. Methods

In computer-aided epilepsy studies, the long-term EEG signal is typically divided into four states: pre-ictal (the period before a seizure), ictal (during a seizure), post-ictal (the short period after a seizure), and interictal (the interval between post-ictal and pre-ictal) [26], [27]. Seizure prediction, therefore, depends on the accurate classification of pre- and interictal EEG signals.

The schematic representation of our proposed MaDin is shown in Fig. 1. MaDin is evaluated with reference to actual clinical patient inclusion. As shown in Fig. 1(a), each subject in the datasets will be specified as the newly admitted patient (target patient). Only one seizure of the target patient is exploited for training together with the samples of other subjects, while the remaining target seizures served as the testing set. After obtaining the raw epileptic EEG signals, the model constructs metastability prior and encodes latent representation with VAE simultaneously. The latent representation is then incorporated into the brain state information through adversarial inference and generalized across subjects through MMD measure. Sections II-A–II-D describe the details of these steps.

A. Dataset Description

This study evaluates performance using two publicly available scalp EEG datasets and one clinical dataset. The CHB-MIT scalp EEG database includes recordings from 23 pediatric patients off anti-epileptic drugs within a few days, sourced on PhysioNet (https://physionet.org/content/chbmit/1.0.0/) [24]. Most data are collected through 22 channels with a sampling rate of 256 Hz, including 844 hours of continuous EEG recording. The Siena dataset contains scalp EEG recordings of 14 adults off anti-epileptic drugs, accessible on PhysioNet (https://physionet.org/content/siena-scalp-eeg/1.0.0/) [25]. Continuous EEG recordings are collected through 29 channels at a sampling rate of 512 Hz.

The clinical dataset includes EEG recordings of seven epilepsy patients recruited at the outpatient ward of Beijing Tiantan Hospital, Capital Medical University. The long-term EEG recordings are collected for pre-operative assessment through 21 channels with a sampling rate of 250 Hz. The study protocol has received approval from the Beijing Tiantan Hospital Institutional Review Board (KY2023-079-01). All patients are provided with written informed consent before participation.

The pre-ictal period is set at 30 minutes, consistent with previous studies [26]. Patients with well-defined seizure types (focal/generalized, if focal: temporal/frontal/occipital) are included. Accordingly, this study utilizes the EEG data of ten qualified patients from the CHB-MIT dataset, six from Siena, and seven from the clinical dataset. Relevant information is presented in Tables V– VII.

B. EEG Preprocessing

For the public available datasets (i.e., CHB-MIT and Siena datasets), the EEG data is filtered using a zero-phase high-pass filter (-3 dB cutoff frequency at 0.5 Hz) and a low-pass filter (-3 dB cutoff frequency at 45 Hz), which excludes the DC component and power line interference, while preserving the major frequency components of epileptic EEG signals [28]. For the clinical dataset, interpolation of the bad channels, and exclusion of the bad segments are followed after filtering. Independent component analysis (ICA) is performed to identify and remove artifactual components related to eye movements,

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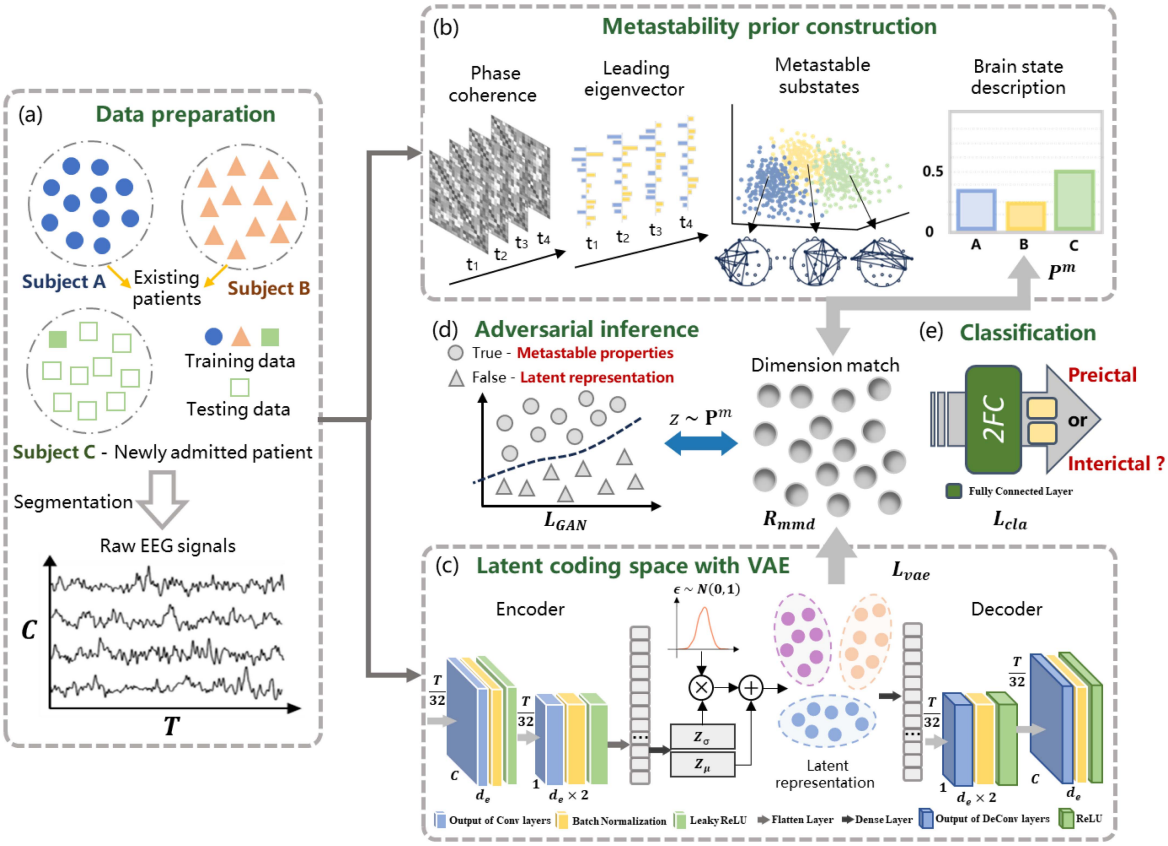

Fig. 1. The schematic representation of our proposed patient-independent seizure prediction model. (a) We begin by dividing the subjects, each of whom will be specified as the newly admitted patient (target subject) and then perform segmentation. (b) We construct metastability prior through four main steps: phase coherence of EEG signal for function connectivity estimation, leading eigenvector generation, modified K-means clustering for metastable substate definition, and brain state descriptions through substate characteristics. (c) A variation autoencoder (VAE) is employed to learn the latent representation in a low-dimensional space while simultaneously preserving the high-order nonlinear information. The network architecture of the encoder consists of convolution layers followed by batch normalization and leaky ReLU activation. The decoder consists of the deconvolution layers followed by batch normalization and ReLU activation. (d) Feature fusion between the learnt latent space and the metastability prior is achieved through adversarial inference, and MMD regularization term R_{mmd} is used to reduce the between-subjects variance. (e) Classification network is connected to the subject-invariant feature space for seizure prediction.

muscle activity, and other non-neural sources. The EEG dataset is finally re-referenced by the average reference. The EEG data is preprocessed with the EEGLAB toolbox (version 2022.0) [29], [30].

The data is then partitioned into pre- and inter-ictal segments based on expert annotations provided with the datasets. The inter-ictal period is much longer than the pre-ictal period, leading to an imbalanced data problem. This study thus performs random subsampling on the inter-ictal sample to make an equal quantity of sample distribution. By setting the sliding window of t-second with 0.5 overlaps, each t-sec EEG segment is represented as a matrix of dimension $(C, f \times t)$, where C denotes the number of channels and f represents the sampling rate, t is set to 20 for all three datasets.

C. Metastability Analysis

To explore the temporal dynamics in the brain network during seizure evolution, a novel framework—leading eigenvector dynamics analysis (LEiDA) [31], [32]—is exploited to analyze the metastability properties of brain networks as shown in Fig. 1(b). LEiDA characterizes brain states as non-random transitions between multiple recurrent metastable substates and quantifies brain states from the probabilistic metastable substate (PMS) space.

Following [31], we utilized instantaneous phase locking to construct the dynamic functional brain network. The phase of EEG signals in all channels $\theta(n_i, t)$ is firstly computed using Hilbert transform [33]. The instantaneous phase locking between each pair of brain regions n_i and n_j at each timepoint t was estimated as the cosine of the relative phase difference as follows:

$$iPL\,(n, p, t) \;= \cos(\,\theta\,(n_i, t) - \theta\,(n_j, t)) \tag{1}$$

phase-locking at a given timepoint ranges from -1 (low phase synchronization) and 1 (high phase synchronization). For each EEG segment, the corresponding iPL is a three-dimensional tensor of size (C, C, T), where C is the number of channels, and T is the number of time points.

The leading eigenvector $V_1(t)$ of each $iPL(t)$ is a $C \times 1$ vector, obtained from the eigenvector corresponding to the largest eigenvalue of the matrix. It reflects the dominant functional connectivity (FC) pattern at timepoint t, which can be reconstructed using the outer product $V_1(t)V_1(t)^T$. Each element

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in $V_1(t)$ represents the projection of the EEG phase of each channel into the leading eigenvector, with positive and negative signs indicating the orientation of the EEG signal. In this regard, the whole brain regions are naturally divided into two clusters, representing the emergence of local sub-networks during the dynamic changes in the brain networks.

To identify recurrent FC patterns across subjects and states, a modified K-means clustering algorithm [34] is applied to divide the set of all eigenvectors of both pre- and inter-ictal states into a predefined number of clusters *k*. The returned *k* cluster centroid vectors V_C with the shape of $C \times 1$ represent the repetitive substates. All leading eigenvectors at each time point will be assigned to the specific cluster through cosine distance measure, generating a discrete state sequence. The optimal number of clusters, meant to metastable substates $S_i \in \{S_1, S_2, \ldots, S_k\}$, is selected using a cross-validation approach on a subset of 3–10 phase-locking patterns. In this study, $k = 5$.

D. The Proposed Patient-independent Seizure Prediction Model (MaDin)

1) Metastability prior construction: To quantify common differences in the functional network from PMS space between pre- and inter-ictal states across subjects, network-level metastable features of discrete state sequence are screened down based on the statistical analysis. Specifically, the sub-sequence corresponding to each t-sec EEG segment is derived from the discrete state sequence obtained via LEiDA. For metastable substates$S_i \in \{S_1, S_2, \ldots, S_k\}$, t_{S_i} and n_{S_i} represent the total duration and the number of the occurrence of S_i, respectively. n_{S_i,S_j} is the number of transitions from S_i to S_j. Four main metastable metrics are calculated for each sub-sequence as follows.

Duration: The average duration after each state occurs.

$$D_{S_i} = \frac{t_{S_i}}{n_{S_i}} \tag{2}$$

Occurrence: The average number of occurrences of each substate per second.

$$O_{S_i} = \frac{n_{S_i}}{\sum_{m=1}^{k} t_{S_m}} \tag{3}$$

Coverage: The proportion of the occurrence time of each substate to the total duration of the sample.

$$C_{S_i} = \frac{t_{S_i}}{\sum_{m=1}^{k} t_{S_m}} \tag{4}$$

Transition probability: The probability of transition between each state.

$$TP(S_i, S_j) = \frac{n_{S_i,S_j}}{\sum_{m=1}^{k} n_{S_i,S_m}} \tag{5}$$

In this regard, forty metastable parameters are obtained, including five coverage parameters, five occurrence parameters, five duration parameters, and 25 transition probability parameters. Permutation-based paired t-test is performed on each parameter to identify significant differences between groups (pre-ictal/interictal). A 10% adaptive False Discovery Rate (FDR) correction based on [35] is used to account for the five metastable substates (S1, S2, S3, S4, and S5, respectively). The sixteen most significant parameters are finally selected to construct the metastability prior, $\mathbf{P}^m$. The overall algorithm of metastability prior construction is described in Algorithm 1.

Algorithm 1: Metastability Prior Construction.

Input: Raw multichannel EEG segments $X \in \mathbb{R}^{N \times C \times T}$
Output: Subject-invariant metastable representation $P \in \mathbb{R}^{N \times 16}$

1: Calculate instantaneous phase locking $iPL \in \mathbb{R}^{C \times C \times t}$ according to (1).
2: Eigenvalue decomposition for each $iPL(\cdot, t_i)$ to obtain the leading eigenvector $V_1(t_i) \in \mathbb{R}^{C \times 1}$.
3: **for** number of clusters $k = 3 \rightarrow 10$ **do**
4: Apply modified *K*-means clustering algorithm to map all eigenvectors $V_1(t)$ into k clusters to generate discrete state sequence.
5: Cross validation to obtain the optimal $k = 5$, the cluster centroid vectors $V_C \in \mathbb{R}^{C \times 1}$ represent the repetitive metastable substates $S_i \in \{S_1, S_2, \ldots, S_k\}$.
6: **end for**
7: Calculate the four main metastable metrics for each discrete state sequence according to (2)-(5).
8: The most significant sixteen metastable matrices are selected as metastability prior $P \in \mathbb{R}^{N \times 16}$ based Permutation-based paired t-test with a 10% adaptive FDR.

2) Latent Coding Space with Adversarial VAE:

a) *Notation:* Suppose there are *M*-1 existing subjects an one target patient in total. The input EEG samples from *M* subjects are denoted by $\mathbf{X}_i = [x_{i_1}, x_{i_2}, \ldots, x_{i_{N_i}}]^{\mathrm{T}}$, where $i \in \{1, \ldots, M\}$, $x_{i_{N_i}} \in \mathbb{R}^{C \times T}$, and N_i is the number of the subject *i*. The one-hot encoding vectors about labels are denoted by $\mathbf{Y}_i = [y_{i_1}, y_{i_2}, \ldots, y_{i_{N_i}}]^{\mathrm{T}}$, where $y_{i_{N_i}} \in \mathbb{R}^{d \times 1}$, and $d = 2$ is the number of classes corresponding to the pre-ictal and ictal state. The metastability prior is denoted by $\mathbf{P}_i^m = [p_{i_1}^m, p_{i_2}^m, \ldots, p_{i_{N_i}}^m]^{\mathrm{T}}$, where $p_{i_{N_i}}^m \in \mathbb{R}^{r \times 1}$, and *r* is set to 16 corresponding to the sixteen most significant metastable parameters described in Section II-D-1. The latent code vectors encoded by the VAE are denoted by $\mathbf{Z}_i = [z_{i_1}, z_{i_2}, \ldots, z_{i_{N_i}}]^{\mathrm{T}}$, where $z_{i_{N_i}} \in \mathbb{R}^{r \times 1}$, and *r* is the dimension of each latent unit equal with the sixteen metastability prior.

b) *VAE:* The VAE, specializing in probabilistic modeling, enforces the latent variables toward a Gaussian distribution for sufficient regularization, enabling the model to generalize better to previously unseen samples [36]. Thus, the adversarial VAE serves as the foundational framework for MaDin.

VAEs typically consist of probabilistic encoder-decoder pairs. The encoder, an inference model $Q_{enc}(x)$, maps a given input *x* into the hidden latent variables *z* as follows:

$$z = \mu + \epsilon \odot \sigma \tag{6}$$

where $\epsilon \sim N(0, 1)$ is an auxiliary noise variable, and $\odot$ indicates the Hadamard product, the mean μ and the standard

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deviation (STD) σ of multivariate Gaussian distribution are generated by the encoder.

Subsequently, the decoder network $P_{dec}(z)$ decompresses the original data—referred to as the reconstructed data $\tilde{x}$—from the latent representation z with the minimum error. The objective loss of VAE is formulated with two parts as (7): the reconstruction loss and the regularization term, given by the Kullback-Leibler (KL) divergence, to force the hidden latent variables approximating the prior Gaussian distribution.

$$L_{vae} = E\left(x - \tilde{x}^2\right) + D_{KL}\left(q_\phi\left(z \mid x\right) \parallel p_g\left(z\right)\right) \tag{7}$$

where $q_\phi(z \mid x)$ is the latent distribution generated by encoder, $p_g(z)$ is the prior distribution following a Gaussian distribution in the latent space, and D_{KL} represents the KL divergence. The reparameterization trick in [37], [38] is employed for estimating D_{KL} as follows:

$$D_{KL} \simeq -\frac{1}{2}\sum_{z=1}^{Z}\left(1 + \log\left(\sigma_z^2\right) - \mu_z^2 - \sigma_z^2\right) \tag{8}$$

where $\mathbf{Z} = \mathbf{16}$ is the dimension of latent space.

a) *MMD-based regularization:* The maximum mean discrepancy (MMD) measure is to learn a feature space shared across patients by minimizing the distribution variance among them. For two arbitrary hidden vectors z_i and z_j from two unseen distributions $\mathbb{P}_i$ and $\mathbb{P}_j$ (also meant to from two patients here), respectively, the instance is mapped to a reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ by adopting the kernel embedding technique [39]. The corresponding mean value in RKHS is given as:

$$\mu\left(\mathbb{P}\right) = \mathbb{E}_{\boldsymbol{z}\sim\mathbb{P}}\ \left[h\left(z, \cdot\right)\right] \tag{9}$$

where $\mu(\cdot)$ is the mean map operation. $h(z, \cdot)$ is the kernel function included by the feature map in $\mathcal{H}$. In this work, the RBF kernel is adopted following the MMD-AAE model [40], [41].

$$\text{mmd}\ \left(\boldsymbol{Z}_i, \boldsymbol{Z}_j\right) = \mu_{\mathbb{P}_i} - \mu_{\mathbb{P}_j\,\mathcal{H}} \tag{10}$$

In this regard, a mmd-based regularization term is given as follows to optimize the hyperparameters in the neural network.

$$R_{\text{mmd}}\ \left(\boldsymbol{Z}^1, \ldots, \boldsymbol{Z}^M\right) = \frac{1}{M^2}\sum_{1\le i,j\le M}\text{mmd}\left(\boldsymbol{Z}_i, \boldsymbol{Z}_j\right) \tag{11}$$

a) *Adversarial feature learning:* VAE combined with MMD regularization effectively extracts subject-invariant feature vectors with complex, high-order nonlinear attributes. To further infuse the underlying epileptic physiological information into the latent space, an adversarial training procedure is utilized to match the aggregated posterior distribution $q(z)$ of the hidden latent codes with the metastability prior $\mathbf{P}^m$.

To be specific, both the latent representation $\mathbf{Z}_i$ and the metastable properties $\mathbf{P}_i^m$ are input into the discriminator $D(\cdot)$. After adversarial inference, the encoder (a generative model) is refreshed by imposing the metastable properties on the data space. The aggregated posterior distribution can fool the discriminator into thinking that the latent representation $q(z)$ originates from the metastability prior. According to [42], the aggregated posterior distribution q(z) on the latent code vector is as follows:

$$q\left(z\right) = \int_x q\left(z \mid x\right) p_d\left(x\right) dx \tag{12}$$

where $q(z|x)$ is the encoding function of the VAE, $p_{\mathrm{d}}(x)$ is the marginal distribution of data.

The objective loss of the adversarial neural network in this work replaces the log-likelihood terms by the least-squared term to relieve the issue of non-convergence [43] as follows:

$$\mathcal{L}_{GAN} = \mathbb{E}_{z\sim p^m}\left[\log D(z)^2\right] + \mathbb{E}_{x\sim p_d}\left[\log\left(1 - D\left(G\left(x\right)\right)\right)^2\right] \tag{13}$$

3) Patient-Independent Seizure Prediction: According to Sections II-D-1 and II-D-2, the EEG-based metastability prior, representing the epileptic characteristics, is first developed. Then, the generalized learnt latent coding space fused with the metastable properties is obtained through VAE, MMD regularization item and adversarial feature learning. The learned epileptic representations are finally used for seizure prediction, that is the classification for preictal and interictal state. The classification procedure employs the cross-entropy loss function, denoted as follows:

$$\begin{aligned}\mathcal{L}_{cla} = &-\frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{n_i=1}^{N_i} y_{i_{n_i}} \cdot \log\left(p\left(y_{i_{\boldsymbol{n_i}}}\right)\right) \\ &+ \left(1 - y_{i_{\boldsymbol{n_i}}}\right) \cdot \log\left(1 - p\left(y_{i_{\boldsymbol{n_i}}}\right)\right)\end{aligned} \tag{14}$$

where $y_{i_{n_i}}$ represents the one-hot encoding vector about label information for supervised learning, and $p\left(y_{i_{\boldsymbol{n_i}}}\right)$ here denotes the predicted class probability.

In sum, MaDin that jointly trains different components is defined as:

$$\min_{Q_{enc}, P_{dec}, C}\max_D \mathcal{L}_{cla} + \lambda_0\mathcal{L}_{VAE} + \lambda_1 R_{mmd} + \lambda_2\mathcal{L}_{GAN} \tag{15}$$

Where λ_0, λ_1, and λ_2 represent the trade-off positive parameters, and C is the classifier.

The training procedure of MaDin consists of two main steps: 1) Training the discriminator to distinguish between hidden codes sampled from the metastability prior and those generated by the encoder, achieved by maximizing the objective (15); and 2) training the VAE to update the encoder and decoder by minimizing input reconstruction error, inter-subject distance, and classification error, achieved by optimizing the encoder Q_{enc}, decoder P_{dec} and classifier C with the same objective. The overall algorithm of MaDin is described in Algorithm 2.

III. Experiment

A. Evaluation Metrics

The seizure occurrence period (SOP) and the seizure prediction horizon (SPH) need to be defined [44] before evaluating prediction performance. SOP is the period in which a seizure is predicted, while SPH is the duration between a seizure alarm and the start of SOP. False alarm arises If an alert is received during SPH but no seizure. In this work, SOP and SPH are set to 30 min and 5 min respectively [45], [46]. We evaluate prediction

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Algorithm 2: Description of the Proposed Patient-Independent Seizure Prediction (MaDin).

Input: Raw multichannel EEG segments $\mathbf{X} = \{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_M\}$, one-hot encoding vectors of corresponding labels $\mathbf{Y} = \{\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_M\}$, initialized parameters Q_{enc}, P_{dec}, C, and D.

Output: Learned parameters Q^*_{enc}, P^*_{dec}, C^* and D^*, and the prediction result of the model.

1: Construct the metastability prior $\mathbf{P}^m$ for $\mathbf{X}$
2: **while** Stopping criterion is not met do
3: Sample a minibatch $\mathbf{X}_b$ and $\mathbf{Y}_b$ from $\mathbf{X}$ and $\mathbf{Y}$, respectively.
4: Sample the corresponding metastable properties $\mathbf{P}^m_b$ from $\mathbf{P}^m$.
5: Compute the gradient of (15) w.r.t. D on $\mathbf{X}_b$
6: Take a gradient step to update D to maximize (15).
7: Compute the gradient of (15) w.r.t. Q_{enc}, P_{dec}, C on $\mathbf{X}_b$, respectively.
8: Take a gradient step to update Q_{enc}, P_{dec}, C to minimize (15), alternatingly.
9: **end while**

performances in terms of sensitivity (SEN), specificity (SPE) and area under the curve (AUC).

As the physiological mechanisms underlying epilepsy remain under active investigation, completely excluding the target patient from training is not feasible. To simulate clinical scenarios, this study incorporates one seizure from the target subject into the training set alongside data from existing subjects, while the remaining seizures from the target subject are reserved for testing. Of the available data, 80% is allocated to the training set, and the remaining 20% is used for validation to monitor overfitting.

B. Experiment Setup

The specific network structures of VAE sub-network in Fig. 1(c), adversarial sub-network in Fig. 1(d) and the classification sub-network in Fig. 1(e) are summarized in Table I. The architecture design of the VAE draws inspiration from the highly effective DeepConvNet [47]. The encoder configuration incorporates temporal convolutions followed by spatial filters, resulting in effective dimensionality reduction. On the other hand, the decoder comprises the corresponding deconvolution layers to reconstruct the original data. The batch normalization and leaky ReLU are applied over the convolution and deconvolution layers. Both the discriminator and the classifier consist of two dense layers that connect the hidden layer to the output categories. To obtain a stable and deterministic input representation, during the test (prediction) phase, the latent representation (z) fed into classifier is the deterministic mean vector (μ) produced by the trained encoder given the input test EEG segment x_{test} [48], that is, $z_{\text{test}} = \mu(x_{\text{test}})$.

This work applies the Adam optimizer to update the parameters of the four modules in (16), where the learning rate is set to 5e-5, and the weight decay is set to 1e-4. The trade-off items conjured by parameter tuning are set $\lambda_0 = 0.03$, $\lambda_1 = 0.03$, and $\lambda_2 = 0.01$. The batch size is set to 32, and the maximum of epochs is set to 50. The experiments are carried out with the Python library and Tensorflow framework on a server equipped with NVIDIA Tesla V100 with 32 GB memory.

IV. Results

Our results are based on the analysis of two publicly available scalp EEG datasets and one clinical dataset described in the Methodology section. We first present the representative results of metastability analysis, then the evaluation of model prediction performance with alation studies of different components, and finally the comparison with other strategies for subject-invariant representation.

A. Metastability Analysis

1) Metastability prior evaluation: To characterize the common metastable patterns, subjects with different seizure types or localizations are analyzed based on the discrete state sequence obtained from LEiDA analysis (described in Section II-C-2). As shown in Fig. 2, there are significant differences for lots of metastable parameters between pre- and inter-ictal states in all three datasets. Specifically, the duration matrices (corresponding to the first row in Fig. 2) for pre-ictal states are significantly lower than that of the inter-ictal. The occurrence matrices (corresponding to the second row in Fig. 2) for pre-ictal states, on the contrary, are significantly higher than that of inter-ictal. The three datasets share the same tendency. It is reasonable to infer that the brain network level during the pre-ictal states tends to be irregular and abnormally active. The coverage matrices (corresponding to the third row in Fig. 2) depend on the total duration of the EEG recording, which varies for each seizure file. The transition probability matrices (corresponding to the fourth row in Fig. 2) indicate the tendency in the transition of brain states to a specific state. Despite statistical significance for most of the four metastable matrices across datasets, both the coverage and transition probability matrices in two states show no consistency probably because of the diversity of seizure types and the differences in the number of each type in the dataset.

2) Metastability findings for patients with specific seizure types: To better analyze the coverage and transition probabilities matrices between these two stages, we conduct group-level analysis for patients with specific seizure types in the dataset. The transition probability matrices for pre- and inter-ictal states are shown in Fig. 3(a) and (b). The transitions with statistical significance are further characterized in Fig. 3(c) and (d). The coverage matrices for pre- and inter-ictal states are shown in Fig. 3(e) and (f).

Transition probability reflects the patterns of constant coupling and decoupling of the brain network, that is, transitions between different metastable substates (coupling modes). A higher transfer probability indicates that the current brain state is more inclined to transition to this coupling pattern. Specifically, there is a significant decrease in the transition probability from other metastable substates to S1, while a significant increase to

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TABLE I
SPECIFIC NETWORK PARAMETERS OF MADIN MODEL

Module	Layer	Input size	Activation	Output size
Encoder	Conv+BN	(~, C, $f \times t$, 1)	LeakyReLU	(~, C, ($f \times t$) // 32, 32)
	Conv+BN	(~, C, ($f \times t$) // 32, 32)	LeakyReLU	(~, 1, ($f \times t$) // 32, 64)
	Flatten	(~, 1, ($f \times t$) // 32, 64)	-	(~,(($f \times t$) // 32) $\times$ 64)
	Dense ($\boldsymbol{Z}_{\mu}$)	(~,(($f \times t$) // 32) $\times$ 64)	-	(~, 16)
	Dense ($\boldsymbol{Z}_{\sigma}$)	(~,(($f \times t$) // 32) $\times$ 64)	-	(~, 16)
	Lamda	(~, 16)	-	(~, 16)
Decoder	Dense	(~, 16)	-	(~,(($f \times t$) // 32) $\times$ 64)
	Reshape	(~,(($f \times t$) // 32) $\times$ 64)	-	(~, 1, ($f \times t$) // 32, 64)
	DeConv+BN	(~, 1, ($f \times t$) // 32, 64)	ReLU	(~, C, ($f \times t$) // 32, 32)
	DeConv+BN	(~, C, ($f \times t$) // 32, 32)	ReLU	(~, C, $f \times t$, 1)
Discriminator	Dense	(~, 16)	ReLU	(~, 256)
	Dense	(~, 256)	Softmax	(~, 2)
Classifier	Dense	(~, 16)	ReLU	(~, 256)
	Dense	(~, 256)	Softmax	(~, 2)

TABLE II
ABLATION EXPERIMENT OF DIFFERENT COMPONENTS

Dataset	Metastable features		No L_{GAN}		No L_{VAE}		No R_{mmd}		MaDin	
	SEN	SPE	SEN	SPE	SEN	SPE	SEN	SPE	SEN	SPE
CHB-MIT	70.9	60.6	90.7	86.9	88.4	91.1	92.9	90.8	93.6	91.6
Siena	61.8.	52.5	81.6	72.5	77.8	81.2	79.4	73.6	88.3	76.3
Clinical dataset	71.0	48.5	78.5	84.7	80.6	77.8	82.7	85.8	87.0	82.6

Metastable features combine with an SVM classifier as a traditional machine learning method without alignment across subjects. No L_{GAN}, removes the adversarial training part described in (16). No L_{VAE}, removes the latent representation encoding with VAE described in (10). No R_{mmd}, removes the MMD-based regularization described in (13).

S3 in patients with temporal lobe seizures. Coverage matrices reflect the current dominant metastable substate to some extent, calculating the proportion of the occurrence time of each substate to the total duration. S1 and S3 are the lowest and the highest in coverage shown in Fig. 3(e), respectively. Moreover, the brain network topology of S1 exhibits stronger functional connectivity/synchronization levels in the temporal lobe brain regions, whereas S3 with stronger functional connectivity in frontal lobe brain regions (shown in Fig. 3(c)). It may suggest that there is some degree of disconnection between the functional brain network of epileptogenic brain regions and other regions during seizure evolution. A similar situation has been found in patients with frontal lobe seizures. During the pre-ictal state, the transition probability both from S1 to S4, and S5 to S4 were significantly decreased (shown in Fig. 3(b)). The coverage of S4 is the lowest shown in Fig. 3(f). The brain network topology of S4 exhibits strong functional connectivity in the frontal lobe, indicating a disconnection between the functional brain network of frontal brain regions and other regions (Fig. 3(d)).

B. Prediction Performance Evaluation

To explore the effect of different components in MaDin on the prediction results, comprehensive ablation experiments on three datasets are performed. For brevity, Table II summarizes the average results for all patients. To rigorously evaluate the contribution of each component, Wilcoxon signed-rank test is conducted based on AUC metrics as shown in Fig. 4(a).

Firstly, the effectiveness of metastability prior is initially examined using metastable features combined with an SVM classifier as a traditional machine learning method without alignment across subjects. Despite the large inter-individual variability, patient-independent prediction using only metastable features can exceed 60% on sensitivity metric, showing the advantage in identifying the epileptic characteristic. This implies the feasibility of further introducing deep learning techniques for a more effective patient-independent seizure prediction model. Besides, our full MaDin model achieves statistically significantly higher AUC compared to this baseline across all three datasets ($p <$ 0.0001 for all three datasets).

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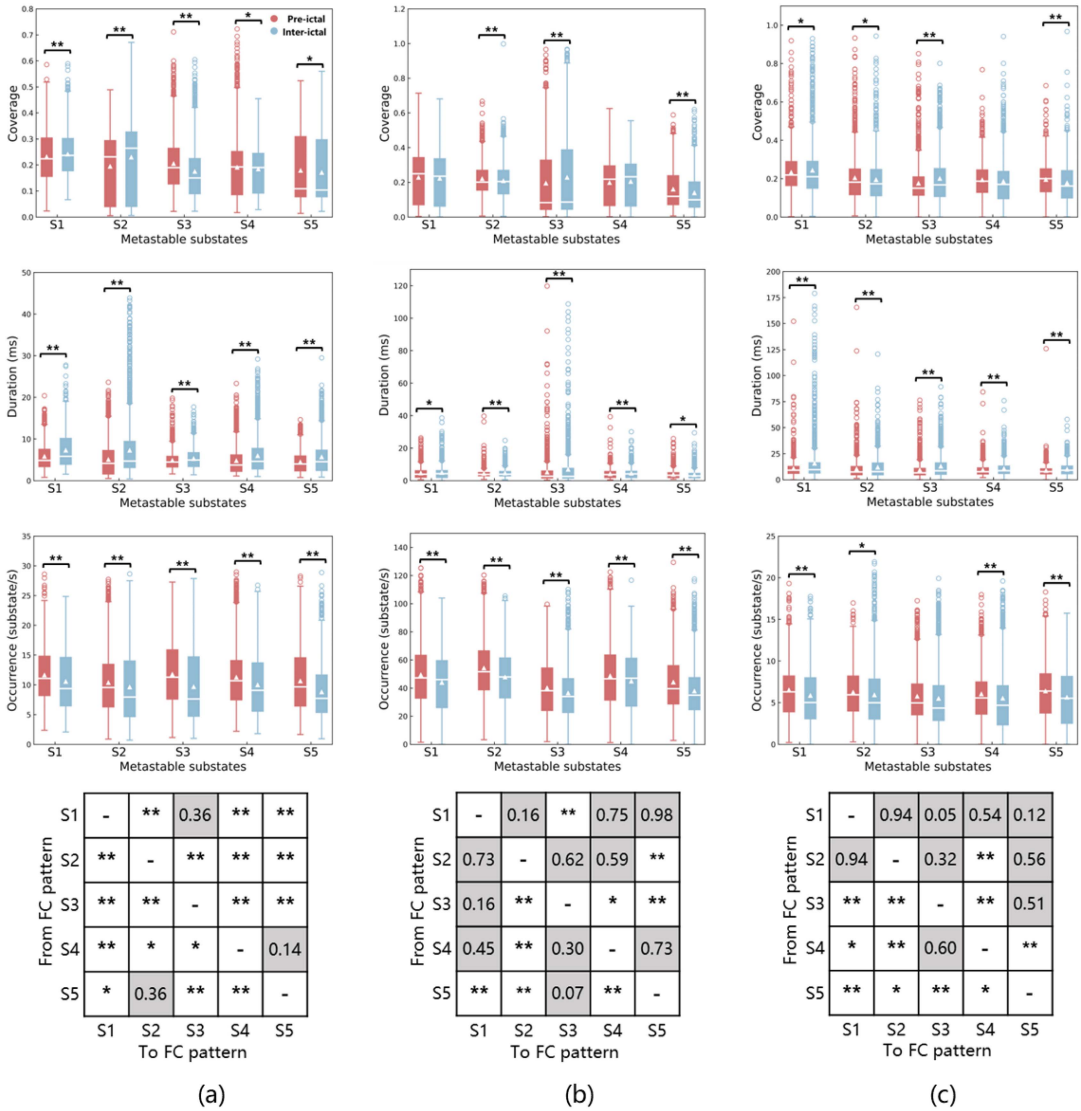

Fig. 2. Distribution of specific characteristics for the metastable substates (S1-S5) for pre- and inter-ictal classes. The first three rows are the metastable matrices of duration, occurrence, and coverage, respectively. The last row is the metastable matric of transition probability. (a)–(c) are the CHB-MIT, Siena, and clinical datasets, respectively. Benjamini and Krieger FDR, q $<$ 0.1, is applied. $*$P $<$0.05; $**$P $<$0.01.

Secondly, each component of the objective function of MaDin described in (16) is examined. From the experimental results, removing each component leads to performance degradation for all three datasets. Removing the adversarial training part (No L_{GAN}) causes the most severe performance degradation in both the sensitivity and specificity metrics. The performance difference in AUC compared to the full MaDin model is statistically significant on the Siena ($p = 0.0219$) and Clinical ($p = 0.0416$) datasets though with no significance on the CHB-MIT dataset ($p = 0.3119$). This underscores the critical role of integrating the metastability prior via adversarial learning.

Replacing VAE with a pure autoencoder (AE) (No L_{VAE}) for the latent representation encoding also results in performance degradation, especially in the sensitivity metric. Similarly, the degradation in AUC when replacing VAE with AE is statistically significant on the Siena ($p = 0.0344$) and Clinical ($p = 0.0342$) datasets. It implies that VAE not only has the advantage of preserving informative and nonlinear attributes epileptic attributes in a highly compressed dimension but also generalizing better to target patient for its extra generative ability compared to AE.

Removing MMD regularization term (No R_{mmd}) causes slight performance degradation in both the sensitivity and specificity metrics in both CHB-MIT and clinical datasets. Statistically, the AUC difference between MaDin and the model without MMD is significant only on the Siena dataset ($p = 0.0012$), while not significant on CHB-MIT ($p = 0.6238$) or Clinical ($p = 0.7168$) datasets. This suggests that combining adversarial VAE and metastability prior for latent representation encoding, using only a small amount of target patient, has already equipped the prediction model with both generalization and discrimination ability.

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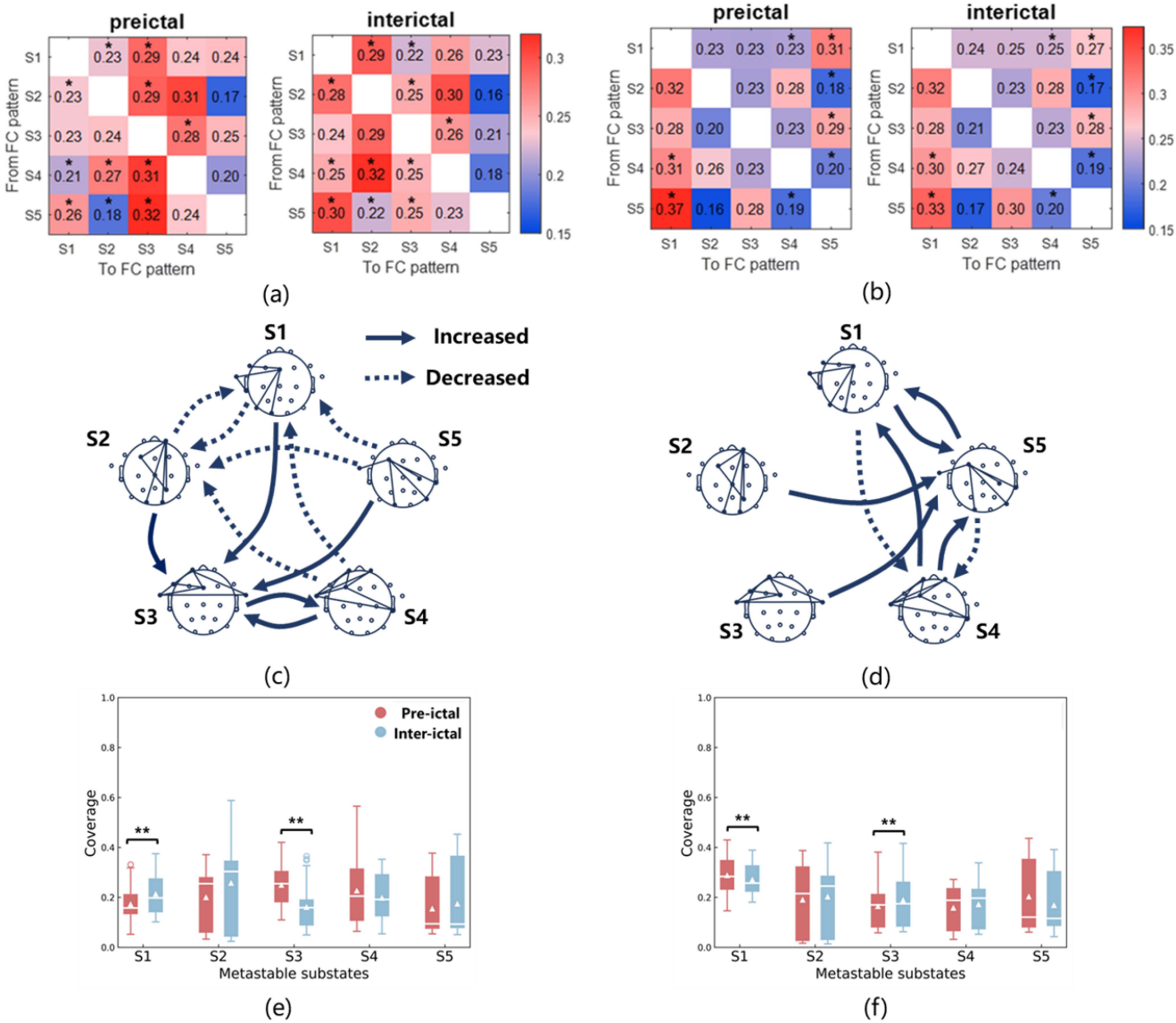

Fig. 3. Transition patterns and coverage matrices for patients with specific seizure types. (a) and (b) represent the transition probability matrices for pre- and inter-ictal states in patients with temporal and frontal lobe seizure types, respectively. The matrices indicate the probability of the transition from a particular substate (row) to any other substate (column). (c) and (d) represent significant transitions ($*P < 0.05$) in patients with temporal and frontal lobe seizure types, respectively. Solid arrows indicate an increased tendency in the transition of brain states to a specific state Si in the pre-ictal phase, whereas dashed arrows indicate a decreased tendency. (e) and (f) represent coverage matrices for patients with temporal and frontal lobe seizure types, respectively.

C. Comparison With Other Strategies for Subject-invariant Representation

Patient-independent seizure prediction studies have referred domain adaptation or generalization techniques to address the inter-subject variability, enabling prediction models to adapt from existing subjects (source domain) to a newly admitted patient (target domain) [23], [40], [49], [50]. This study further compares three main strategies for subject-invariant representation. All the experiments are conducted in a unified setting. A detailed description of these methods is presented below.

DAN uses a multi-kernel MMD (MK-MMD) to replace the original single-kernel MMD for image classification, which aims to solve the problem of the restricted kernel function of MMD [51].

SIDA learns invariant representations via adversarial learning for motor imagery tasks [52]. Adversarial representation learning in this work is viewed as simultaneously learning to predict a dependent variable from a representation while exploiting an adaptive dependence measure between these two to also learn the representation itself such that this dependence is minimized.

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CHB-MIT dataset

Methods	MS	No L_{GAN}	No L_{VAE}	No R_{mmd}	MaDin
MS	-	<0.0001	<0.0001	<0.0001	<0.0001
No L_{GAN}	<0.0001	-	0.6813	0.4273	0.3119
No L_{VAE}	<0.0001	0.6813	-	0.7142	0.4121
No R_{mmd}	<0.0001	0.4273	0.7142	-	0.6238
MaDn	<0.0001	0.3119	0.4121	0.6238	-

Methods	DAN	SIDA	MMD-AAE	MaDin
DAN	-	0.0225	0.0020	0.0020
SIDA	0.0225	-	0.0695	0.3500
MMD-AAE	0.0020	0.6953	-	0.4316
MaDin	0.0020	0.3500	0.4316	-

Siena dataset

Methods	MS	No L_{GAN}	No L_{VAE}	No R_{mmd}	MaDin
MS	-	<0.0001	<0.0001	<0.0001	<0.0001
No L_{GAN}	<0.0001	-	0.8672	0.2487	0.0219
No L_{VAE}	<0.0001	0.8672	-	0.6731	0.0344
No R_{mmd}	<0.0001	0.2487	0.6731	-	0.0012
MaDn	<0.0001	0.0219	0.0344	0.0012	-

Methods	DAN	SIDA	MMD-AAE	MaDin
DAN	-	0.6684	0.2333	0.0004
SIDA	0.6684	-	0.0692	0.0047
MMD-AAE	0.2333	0.0692	-	<0.0001
MaDin	0.0004	0.0047	<0.0001	-

Clinical dataset

Methods	MS	No L_{GAN}	No L_{VAE}	No R_{mmd}	MaDin
MS	-	<0.0001	<0.0001	<0.0001	<0.0001
No L_{GAN}	<0.0001	-	0.7462	0.0614	0.0416
No L_{VAE}	<0.0001	0.7462	-	0.1072	0.0342
No R_{mmd}	<0.0001	0.0614	0.1072	-	0.7168
MaDn	<0.0001	0.0416	0.0342	0.7168	-

Methods	DAN	SIDA	MMD-AAE	MaDin
DAN	-	0.0006	0.0002	<0.0001
SIDA	0.0006	-	0.7401	0.9545
MMD-AAE	0.0002	0.7401	-	0.8742
MaDin	<0.0001	0.9545	0.8742	-

(a) (b)

Fig. 4. Wilcoxon signed-rank test between pairing methods based on AUC metric in the three datasets at the significance level of 0.05. (a) Statistics results for ablation experiments. (b) Statistics results for different subject-invariant representation methods. Statistical difference is significant if $p < 0.05$ shaded with gray.

TABLE III
PREDICTION PERFORMANCE COMPARISON WITH EXISTING METHODS FOR SUBJECT-INVARIANT REPRESENTATION

Dataset	DAN [51]		SIDA [52]		MMD-AAE [53]		MaDin	
	SEN	SPE	SEN	SPE	SEN	SPE	SEN	SPE
CHB-MIT	73.6	68.9	85.8	88.5	88.2	91.6	**93.6**	**91.6**
Siena	68.0	54.5	68.5	58.5	60.5	51.9	**88.3**	**76.3**
Clinical	50.2	60.0	80.2	**88.0**	86.3	84.5	**87.0**	82.6

The best result for each column (SEN and SPE) is marked in bold.

MMD-AAE first extends adversarial autoencoders for object recognition by imposing the MMD measure to align the distributions among different domains and matching the aligned distribution to an artificial Laplace prior distribution through adversarial feature learning [41].

For brevity, Table III summarizes the average results for all patients. The patient-by-patient results are summarized in Tables VIII–X. The Wilcoxon signed-rank test is conducted based on AUC metrics as shown in Fig. 4(b). DAN gives the worst results compared to other methods, only focusing on the alignment of domain marginal distribution via MK-MMD. The alignment potentially destroys the discriminative hyperplane of the features due to the excessive gap between different patient domains, resulting in poor prediction performance. Both SIDA and MMD-AAE significantly outperform DAN and achieve comparable performance. SIDA verifies the effectiveness of adversarial training on generalizability. Though MMD-AAE constructs an artificial Laplace prior, the performance is not significantly improved as expected.

Compared with the three methods, MaDin achieves the best performance in both sensitivity and specificity metrics on almost all three datasets. Statistical analysis of the AUC metric further confirms the superiority of MaDin, showing significantly better performance than DAN across all datasets ($p < 0.05$) and significant advantages over SIDA and MMD-AAE specifically on the Siena dataset ($p < 0.05$). Though AUC improvement of MaDin over SIDA and MMD-AAE has not reached significance on CHB-MIT and Clinical datasets, MaDin exhibits greater performance stability across individual subjects. We have not observed the severe prediction failures seen with comparison methods in certain cases, such as the sensitivity/specificity of SIDA for subject chb02 (40% /90%) or MMD-AAE for subject chb18 (44% /97%) on the CHB-MIT dataset, suggesting a potential lack of capability in recognizing pre-ictal features for these specific instances by the comparison models. Similar instances of inconsistent performance for SIDA and MMD-AAE are observed on the clinical dataset as well. The consistent performance of MaDin across subjects suggests higher reliability, an important factor for clinical translation.

V. DISCUSSION

In this study, we develop an EEG-based epileptic representation by fusing the underlying neurological process of epileptic brain network with the latent coding space, enabling reliable patient-independent seizure prediction. Our results have demonstrated that our prediction model, MaDin, achieved the best prediction performance among other state-of-the-art patient-independent seizure prediction algorithms with the same subject division for validation. The brain-network-inspired correspondence and reliable prediction performance show the advantages of our invariant representation integrated with network temporal dynamics.

A. Investigation of EEG-based Metastability Findings

The spiking during the inter-ictal state indicates a more deterministic activity, while the ictal state or the pre-ictal state tends to be a more complex and chaotic activity, and spectral chirps in EEG [54]. Moreover, when seizure approaches, fluctuation in neural signals near the seizure onset area, far from equilibrium, exhibits synchronized instabilities in avalanche dynamics [55]. Comparably, as described in Section IV-A-1, we first observed that compared to the inter-ictal states, the occurrence of five metastable substates increased in the pre-ictal states, while their corresponding durations decreased, reflecting a highly disordered state of irregular transitions between different metastable substates in the pre-ictal state (up to 30 minutes before epileptic seizures).

Based on the mentioned above, we may naturally expect to further analyze whether the differences in metastable patterns between the pre- and inter-ictal can be observed in the different seizure types. To this end, we further conduct a detailed analysis for the patients with specific seizure types as shown in Fig. 3.

Seizures destabilize the brain networks. A critical transition arises from normal brain activities to the ictal states [56]. "Critical slowing down" is considered as a typical phenomenon when approaching the critical transition [57]. Additionally, epileptic

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TABLE IV
PREDICTION PERFORMANCE COMPARISON WITH EXISTING PATIENT-INDEPENDENT METHODS ON CHB-MIT DATASET

Method	Strategy	SPH-SOP (min)	AUC (%)	SEN (%)	SPE (%)
MMD-AAE-SP [40]	MMD regularization, Laplace prior based adversarial training	-30	75.0	73.0	76.0
MAAE [23]	Riemannian manifold prior based adversarial training	0-30	84.0	82.0	87.0
SSDA-SPM [49]	MMD regularization, consistency regularization	5-30	84.9	88.8	81.8
AdaTrans [50]	Distiller, conditional adversarial training	5-30	81.4	79.5	74.2
MaDin	Metastability prior based adversarial training	5-30	**94.2**	**93.6**	**91.6**

The best result for each column (AUC, SEN and SPE) is marked in bold.

network shows transiently disrupted balance with long-range connectivity with other networks [58], which is an important process for the generation of epileptic activity. In this study, while analyzing the patients with temporal lobe seizures as shown in Fig. 3(c), if the topological structure distribution of specific metastable substates primarily involves the temporal lobe structures, the transition "slowing" of these subnetworks into other substates will occur when impending seizure onset, alternatively seen as "being functional isolated". The corresponding phenomenon still exists in frontal lobe epilepsy shown in Fig. 3(d).

Taken together, (i) metastable substates overlapping with the seizure onset regions exhibit decreased transition probabilities in the pre-ictal states; (ii) in contrast, for those sub-states whose primary spatial domain does not involve the seizure onset regions, its coverage matric plays a predominant role in the pre-ictal periods shown in Fig. 3(e) and (f). According to the variations among metastable substates existing in the pre- and inter-ictal states, we infer that decreased transitions towards the substates associated with focal seizure areas, and the increased reorganization with substates that are distant from seizure sources, may reflect a progressive loss of the ability to maintain regional stability, and also an increase in the spread of coupling to other regions before catastrophic events [59].

B. The Advantages of MaDin for Patient-independent Seizure Prediction

Table IV summarizes all the studies for patient-independent seizure prediction. All studies were validated in the same way as ours. The training set includes the data of existing subjects and one seizure from the target subject, while the remaining seizures of the target subject are only for testing. There are still relatively few studies under such validation for patient-independent seizure prediction due to the well-acknowledged inter-patient variability. Most of these studies have validated their methods only on the CHB-MIT dataset. As shown in Table IV, MMD-AAE-SP and MAAE first explored this challenging issue [23], [40]. MMD-AAE-SP, derived from MMD-AAE [41], preprocessed the raw EEG signals with a short-time Fourier transform, essentially converting epileptic EEG signal classification into an image classification problem. Both MMD-AAE-SP and MAAE followed the learning strategy of autoencoder to map the input data into high-dimensional embeddings. To generalize well on the target patient, the artificial prior, such as Laplace prior or Riemannian manifold-based prior, is imposed to the aligned latent space through adversarial feature learning. Riemannian manifold-based prior outperforms by capturing spatially correlated features from EEG compared to artificial Laplace priors without task-specific information. Both Riemannian manifold-based prior and our proposed metastability prior serve as feature extraction for raw EEG signals [60]. When combined with an AE for latent space encoding, MAAE incorporates information from only a single seizure event of the target subject into the encoding process. However, it is less able to effectively mitigate overfitting on the existing subject data without additional generalization techniques. This is the key factor why we employ a VAE instead of AE to generate the latent space. Nor have MAAE defined SPH for adequate clinical interventions. To further improve cross-subject generalizability, feature alignment strategies that minimize the embedding distribution disparity between every two patients, and semi-supervised techniques that utilize the unlabeled data of target patients for finetune, have also been explored [49], [50]. These studies have had little or no effect on performance improvement, which suggests that directly adopting classical domain-related strategies in image classification and speech recognition cannot further adapt to EEG representation learning.

The basic idea of the above studies is to identify robust common representation across subjects. Our proposed framework, MaDin, first characterizes the temporal dynamics of epileptic brain networks through metastability analysis. The characteristics of the metastability prior, which show consistency with the physiological mechanism of epilepsy, provide a physiologically grounded foundation for representation learning, moving beyond purely statistical pattern matching or the use of task-agnostic priors. By utilizing the generative capability of VAE and adversarial training, the domain knowledge is integrated into the latent space, further regularized by MMD regularization. This knowledge-guided idea ensures that the learned subject-invariant representation preserves not only feature discriminability for prediction but also generalization across both existing subjects and the target subject (the newly admitted patient). Compared with existing methods, MaDin obtains a benefit on the CHB-MIT dataset of AUC, sensitivity, and specificity by about 9%, 5%, and 5%, respectively.

Finally, beyond the advantages in predictive accuracy and robust representation learning, MaDin demonstrates significant practical advantages in terms of computational efficiency. As shown in Fig. 5, our analysis revealed very low inference times (sub-second for processing batches of 120 samples, translating to millisecond-level per-segment speed) which are statistically consistent across all tested datasets (Kruskal-Wallis, $p > 0.05$). This high efficiency strongly supports the feasibility of

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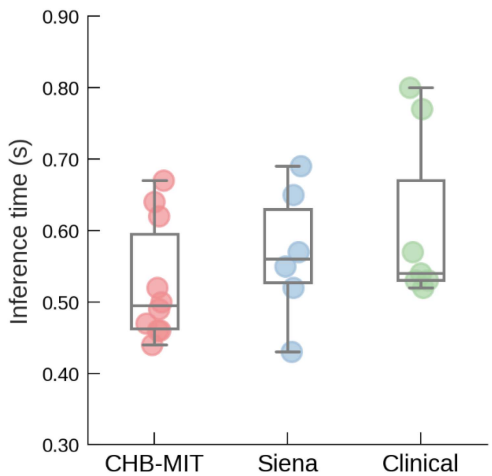

Fig. 5. Inference time for the three datasets. Each scatter represents a subject. No significant difference among the three datasets with Kruskal-Wallis H test ($p > 0.05$).

deploying MaDin for real-time clinical applications [61], complementing its state-of-the-art predictive performance.

C. Limitations and Future Work

While our MaDin model demonstrates promising patient-independent seizure prediction by integrating a metastability prior, we acknowledge limitations and outline future directions. The current validation is based on specific pediatric and adult scalp EEG datasets; broader validation across larger populations, diverse seizure types, and multi-center data are necessary. Our focus on segment-based evaluation metrics, while fundamental, requires developing post-processing techniques, especially parameter-free, to assess event-based clinical metrics like false positive rate per hour (FPR/h) within the defined SPH framework in the future. Besides, detailed failure case analysis, which is crucial for understanding model weaknesses, has not been included.

Future research will prioritize these aspects: performing broader validation, developing event-based evaluation protocols with post-processing techniques, and conducting in-depth failure analyses. Additionally, we aim to explore more adaptive methods for capturing pre-ictal dynamics and investigate deeper integration of physiological knowledge to further enhance model robustness, generalizability, and interpretability. Addressing residual inter-subject heterogeneity and optimizing clinical parameters like SPH/SOP also remain important considerations for translating this work towards reliable clinical application.

VI. Conclusion

Exploring effective integration between the underlying neural processes specific to epilepsy, and the deep learning models commonly criticized for black-box problems, is one of the most challenging tasks in computer-aided epilepsy studies. To this end, we propose a metastability-based deep learning framework to construct epileptic representations with network temporal dynamics, enabling reliable patient-independent seizure prediction. The generalization ability and prediction performance of our proposed MaDin have been evaluated on the two publicly available scalp EEG datasets and one clinical dataset. Owing to the promising experimental results of MaDin, the integration of metastable properties with deep learning techniques deserves further in-depth investigation to achieve better interpretability and more accurate patient-independent seizure prediction.

Appendix

TABLE V
Detailed Description of CHA-MIT Dataset

Subject	Age	Gender	No. of seizures	Seizure type	Localization
chb 1	11	F	7	SP, CP	Frontal
chb 2	11	M	3	SP, CP, GTP	Temporal
chb 3	14	F	7	SP, CP	Frontal
chb 5	7	F	5	CP, GTC	Frontal
chb 10	3	M	7	SP, CP, GTC	Temporal
chb 17	12	F	3	CP, GTC	Temporal
chb 18	18	F	6	SP, CP	Frontal
chb 19	19	F	3	SP, CP, GTC	Frontal
chb 22	9	F	3	SP, CP	Temporal
chb 23	6	F	7	SP, CP	Temporal

F, female; M, male; SP, simple partial; CP, complex partial; GTC, generalized tonic-clonic

TABLE VI
Detailed Description of Siena Dataset

Subject	Age	Gender	No. of seizures	Seizure type	Localization
PN03	54	M	2	CP	Right Temporal
PN06	36	M	5	SP	Left Temporal
PN07	20	F	1	SP	Left Temporal
PN12	71	M	4	SP, GTC	Left Temporal
PN14	49	M	4	CP, GTC	Left Temporal
PN16	41	F	2	CP	Left Temporal

TABLE VII
Detailed Description of Clinical Dataset

Subject	Age	Gender	No. of seizures	Seizure type	Localization
Pt 1	24	M	3	CP	Left Frontal
Pt 2	36	M	2	SP	Right Frontal
Pt 3	18	F	2	SP	Right Parietal
Pt 4	24	F	2	SP, GTC	Left Parietal
Pt 5	17	F	2	CP, GTC	Right Temporal
Pt 6	11	F	4	CP	Right Temporal
Pt 7	21	F	2	SP	Right Frontal

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TABLE VIII
PREDICTION PERFORMANCE COMPARISON WITH EXISTING METHODS FOR SUBJECT-INVARIANT REPRESENTATION ON CHB-MIT DATABASE

Target	DAN [51]		SIDA [52]		MMD-AAE [53]		MaDin	
	SEN	SPE	SEN	SPE	SEN	SPE	SEN	SPE
chb1	83.7	80.6	92.3	87.7	94.7	75.7	81.7	97.9
chb2	90.9	76.7	40.0	95.0	95.2	98.3	97.1	91.7
chb3	69.3	81.2	98.3	98.3	95.2	86.3	94.6	94.3
chb5	84.4	29.4	82.6	52.5	78.3	95.8	85.6	76.4
chb10	54.5	74.0	98.3	74.0	95.4	91.9	89.9	81.1
chb17	74.3	41.8	99.4	99.4	97.2	96.4	100.0	90.6
chb18	53.5	60.1	77.0	88.9	44.1	97.0	93.0	93.2
chb19	97.3	87.5	99.1	95.8	92.5	96.7	100.0	99.0
chb22	93.6	75.1	73.2	99.2	99.2	90.8	100.0	95.4
chb23	34.9	82.3	97.7	93.7	100.0	87.5	94.1	96.6
Avg.	73.6	68.9	85.8	88.5	88.2	91.6	93.6	91.6

TABLE IX
PREDICTION PERFORMANCE COMPARISON WITH EXISTING METHODS FOR SUBJECT-INVARIANT REPRESENTATION ON SIENA DATABASE

Target	DAN [51]		SIDA [52]		MMD-AAE [53]		MaDin	
	SEN	SPE	SEN	SPE	SEN	SPE	SEN	SPE
PN03	81.1	52.4	78.9	79.8	67.7	54.6	96.3	76.6
PN06	73.4	46.3	43.2	70.1	89.8	20.7	91.2	69.5
PN07	35.4	87.0	80.0	67.8	54.8	77.8	85.4	88.6
PN12	81.4	62.9	91.8	37.0	62.2	45.3	92.7	80.1
PN14	81.7	14.3	63.9	54.8	56.2	47.7	87.1	68.9
PN16	54.9	64.4	53.4	41.8	32.2	65.5	76.9	74.0
Avg.	68.0	54.5	68.5	58.5	60.5	51.9	88.3	76.3

TABLE X
PREDICTION PERFORMANCE COMPARISON WITH EXISTING METHODS FOR SUBJECT-INVARIANT REPRESENTATION ON CLINICAL DATABASE

Target	DAN [51]		SIDA [52]		MMD-AAE [53]		MaDin	
	SEN	SPE	SEN	SPE	SEN	SPE	SEN	SPE
Pat1	26.4	75.2	76.1	68.4	70.9	76.1	85.7	89.7
Pat2	38.0	76.6	62.8	75.0	63.2	90.2	90.4	79.2
Pat3	71.5	44.1	96.8	89.5	98.8	80.4	92.6	81.8
Pat4	35.7	65.8	95.5	99.3	86.6	99.1	82.7	81.7
Pat5	89.0	23.6	88.8	95.5	87.1	96.6	84.8	77.3
Pat6	49.5	67.3	61.4	100.0	97.5	87.0	84.8	84.3
Pat7	41.2	67.3	90.4	92.3	100.0	62.2	88.3	84.4
Avg.	50.2	60.0	80.2	88.0	86.3	84.5	87.0	82.6

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